Whom to Trust? Trust-Aware Delegation for Decentralized LLM-Agent Networks

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1 Introduction

PPAI [1] lets each user's edge agent delegate a query to a peer whose agent is more capable. Queries and agents are matched in a shared prototype space, and a Bayesian load game picks a capable, lightly loaded peer. PPAI, however, *trusts every peer*: capability vectors are self-reported and answers are accepted at face value. In our simulator, this makes it fragile. With 10% of peers lying about their skill, PPAI's accuracy falls from 0.73 to 0.19; with 10% colluding on a shared wrong answer, it falls to 0.04. We study *trust-aware delegation*: whom to ask, how many peers to ask, and how to combine their answers when some peers are adversarial and every verification costs a network round trip. So far we have built a simulator grounded in PPAI's measured accuracies, together with four baselines. We have also built TADA (Trust-Aware Delegation with Audits) and a first Q-learning controller. With 30% colluders, TADA keeps 0.77 accuracy, where Dawid–Skene and majority voting fall to $\leq$0.13 and EigenTrust reaches 0.64, and it sends 17% fewer probes.

2 Problem Statement and Formulation

System. There are n peers. Each query belongs to one of K skill clusters (PPAI's prototypes). Peer j advertises a capability vector $r_j \in [0,1]^K$ and answers a cluster-k query correctly with an unknown probability a_{jk}. Each peer is a FIFO server with rate μ_j, and a remote probe adds a delay d in each direction. A query q has a cluster k_q and a true answer $y_q \in \{1,\dots,L\}$ (multiple choice, $L = 4$). **Decision.** The owner's policy π sends probe sets $S_1, S_2, \dots$ in rounds and waits for each round's answers before deciding whether to stop. It then outputs $\hat{y}_q$. It may also send *audits*, i.e. gold questions with known answers. The objective is

$$\max_\pi \ \mathbb{E}\big[\mathbb{1}\{\hat{y}_q = y_q\}\big] - \gamma\,\mathbb{E}\big[|S(q)| + A(q)\big] - \eta\,\mathbb{E}[T_q], \tag{1}$$

where $|S(q)|$ counts probes, $A(q)$ counts audits, and T_q is the end-to-end latency. There is no trusted third party, no pre-trusted peer, and no ground truth beyond the audit pool. **Threats.** A fraction f of peers advertise $r_{jk} = 0.95$ everywhere: (T1) *skill liars* answer correctly with probability 0.15; (T2) *colluders* all give the same wrong answer to each query; (T3) colluders *whitewash*, re-joining under a fresh identity when their reputation drops below 0.4 [2]; (T4) colluders *camouflage* by answering correctly with probability 0.3. For now, gold questions are assumed indistinguishable from user queries. **Why AI.** Peer reliability is latent and must be inferred from noisy, correlated answers (Bayesian inference, EM). Stopping is a sequential decision, i.e. a small MDP for reinforcement learning. Choosing whom to audit is an exploration–exploitation (bandit) problem.

3 Motivation

Interesting: P2P trust is well studied for file sharing, where a download can be checked against a hash. LLM answers have no hash. Honest peers are imperfect (accuracies of 0.13–0.87 in PPAI's Table I) and their errors are correlated, so disagreement is weak evidence of malice, while a coalition can manufacture agreement. **Important:** open agent-to-agent networks are emerging, a wrong medical or financial answer is costly, and PPAI lists trust as an open problem. Verification also consumes network resources: each extra probe costs a round trip and GPU time. Deciding how much to verify is therefore a networking resource-allocation problem, not only a security problem.

4 Related Work

LLM routing and ensembles. RouterDC [3], RouteLLM [4] and KABB [5] route queries on one trusted platform. Self-consistency [6] and "More Agents" [7] show that aggregating answers helps, but they assume honest members. PPAI [1] decentralises routing but keeps that assumption. **P2P reputation.** EigenTrust [8] anchors global trust on pre-trusted peers. PeerTrust [9] and Beta reputation [10] aggregate feedback about transactions, and surveys [11, 12] catalogue whitewashing and Sybil attacks [13]. All of these assume transaction outcomes are observable. **Crowdsourced truth inference.** Dawid–Skene [14], GLAD [15] and Karger et al. [16] infer worker reliability without labels, but assume independent errors, which collusion breaks. Gold questions [17] inspire our audits. **BFT and verifiable inference.** BFT [18, 19] needs deterministic replicas, whereas honest LLMs legitimately disagree. zkLLM [20] is far too slow for interactive queries. **Gap:** we are not aware of work that combines latent-reliability inference with adaptive probe allocation under collusion in a delay-sensitive network.

5 Plan of Action

When	Milestone	Lead
Oct 13–21	Simulator, baselines, TADA, first RL controller *(done)*	Both / Adele
Oct 22–28	Milestone 2 report and progress slides	Both
Oct 29–Nov 11	Real answers of the 5 models (vLLM [21]); real traces [22, 23]	Trilok
Nov 12–25	Better learned control (per-round credit, audit bandit); stronger attacks	Adele
Nov 26–final	Final experiments, IEEE final report ($\geq$6 pages), 25-min talk	Both

Division of work. T. Padhi proposed the core direction of modelling peer disagreement to assign trust. He leads threat models, the literature review, data pipelines and evaluation. A. Chinda leads the simulator, TADA and the RL controller. Both of us write and present.

6 Research Progress

Simulator and data. We built a discrete-event simulator with Poisson arrivals, FIFO peer queues and network delay. Policies are written as processes that send probes and wait for the replies. It has 50 peers, each an instance of one of PPAI's five agents (Qwen2.5-7B, DeepSeek-R1-Distill-7B, Nemotron-Code-7B, HuatuoGPT-o1-7B, Llama-3-8B). Their accuracies on MMLU, GSM8K, MedQA, ARC-C and AGIEval are taken from PPAI's Table I, plus per-peer noise. Honest errors fall on a shared distractor with probability 0.5, so they are correlated. Other settings: service rates of 15–25/s, a one-way delay of 5 ms, an arrival rate $\lambda = 50$/s, 5000 queries per run, and 5 seeds. **Baselines:** PPAI's argmax; majority vote over the 5 most "relevant" peers; Dawid–Skene EM; and EigenTrust, which we give 3 pre-trusted honest peers (TADA gets none).

TADA. Per query, it (1) shortlists peers by reported relevance (PPAI's scoring stage); (2) ranks them by *estimated* accuracy $\hat{a}_{jk}$ minus a load penalty, so lying about skill buys nothing; (3) asks the top 3 and, if the posterior of the best answer is below $\tau = 0.9$, asks 2 more; and (4) with probability 0.15, sends a gold question from PPAI's own per-prototype question set Q_k to a peer it just used or to the peer whose audit record is most uncertain. Each peer keeps Beta counts per cluster [10] for agreement, $(\alpha_{jk}, \beta_{jk})$, and for audits, (g^+_{jk}, g^-_{jk}). Agreement may never claim more than the audit record supports:

$$\hat{a}_{jk} = \min\left\{\tfrac{\alpha_{jk}}{\alpha_{jk}+\beta_{jk}},\, U(g^+_{jk}, g^-_{jk}),\, U(g^+_{j\cdot}, g^-_{j\cdot}) + \delta\right\}, \quad U = \hat{p} + z\sqrt{\hat{p}(1-\hat{p})/(n+1)}, \qquad (2)$$

where $\hat{p}$ and n are the audit success rate and audit count, and the last term pools a peer's audits over all clusters. Answers are combined with the one-coin posterior $P(y{=}\ell \mid x_S) \propto \prod_{j\in S} \hat{a}_{jk}^{\mathbb{1}[x_j=\ell]} \left(\frac{1-\hat{a}_{jk}}{L-1}\right)^{\mathbb{1}[x_j\neq\ell]}$. **Anchored agreement:** agreement counts are updated only if the same answer still wins (posterior ≥ 0.5) when each peer is weighted by its *audit-only* lower bound, floored at chance. Unaudited colluders then carry no weight and cannot bootstrap their own trust.

Initial findings. (1) PPAI is captured by any liar. Dawid–Skene handles independent liars but collapses under collusion (0.01), because colluders look like a reliable majority (Fig. 1a,b). (2) TADA

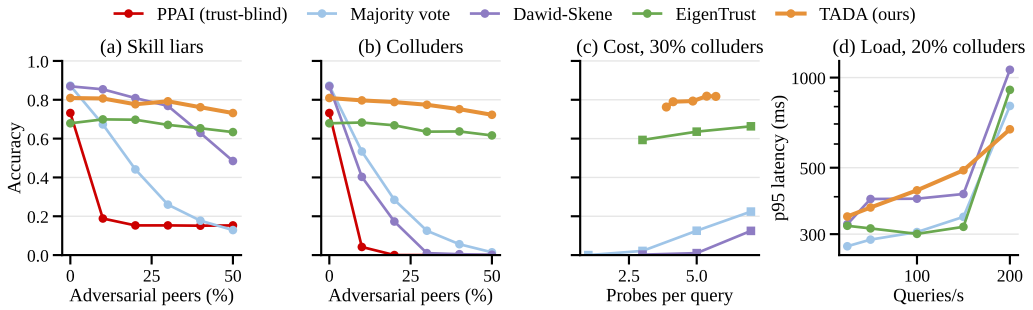


Figure 1: (a,b) Accuracy as the share of adversarial peers grows. (c) Accuracy vs. probes per query (committee size or τ swept). (d) p95 latency vs. arrival rate. Means over 5 seeds.

Table 1: Accuracy with 30% adversaries of each kind; probes and p95 latency are measured with colluders. The rows marked − remove one TADA component.

Method	No attack	Liars	Colluders	+Whitewash	+Camouflage	Probes	p95 (ms)
TADA (full)	0.81	**0.79**	**0.77**	**0.75**	**0.76**	4.2	376
− anchoring	0.82	0.78	0.63	0.47	0.67	4.1	370
− audits	0.76	0.57	0.42	0.42	0.61	4.8	432
− pooled audit cap	0.81	0.79	0.76	0.74	0.76	4.2	373
− sequential (ask 5 at once)	0.82	0.79	0.77	0.75	0.76	5.1	276
EigenTrust (m=5)	0.68	0.67	0.64	0.67	0.66	5.0	301
Dawid–Skene (m=5)	**0.87**	0.77	0.01	0.01	0.34	5.0	333
Majority vote (m=5)	**0.87**	0.26	0.13	0.13	0.31	5.0	268

reaches 0.77 with 30% colluders and 0.72 with 50% (EigenTrust: 0.64 and 0.62), without any pre-trusted peers. Only 4% of the answers it uses come from adversaries, and it survives whitewashing and camouflage. Accuracy climbs from 0.53 to 0.83 within the first 30% of a run. (3) Audits and anchoring do the work (Table 1): without audits, accuracy under collusion is 0.42; without anchoring, 0.63 (0.47 under whitewashing). The pooled audit cap adds little. (4) Verification costs the network. Sequential rounds save 18% of probes but raise p95 latency from 276 to 376 ms. Under heavy load the saved probes pay off: at 200 queries/s, TADA has the lowest p95 latency (672 ms vs. 804–1062 ms; Fig. 1d). (5) The price of scepticism: with no attack, TADA is 6 points below a full 5-peer vote. **Preliminary RL.** Tabular Q-learning [24] replaces τ. The state is (confidence, round, load, cluster), the actions are {stop, ask 2 more}, and the reward is correctness minus costs per probe and per second of latency. It is trained in simulation on seeds separate from the evaluation seeds. With 30% colluders at 150 queries/s, it learns to stop earlier: p95 latency falls from 576 to 310 ms and probes from 4.3 to 3.3, for −1.4 accuracy points. Without the latency penalty it matches the best threshold (0.79) but does not beat it; binary rewards are noisy.

Challenges and how we addressed them. (a) Our first answer model never let wrong answers agree, which made voting look almost perfect. We switched to a shared-distractor multiple-choice model. (b) Agreement-only trust was captured by colluders (0.003 accuracy). Capping trust by audits fixed the steady state, and anchoring fixed the warm-up (+14 points). (c) Our first queue model reserved servers for future verification rounds, producing spurious 12 s latencies. We rebuilt the simulator as a discrete-event simulation. (d) Asking one peer first caused up to 3 sequential rounds. Asking 3 first caps a query at two rounds and cuts p95 latency by about 25%. **Next algorithms:** per-round credit assignment for the RL controller; a contextual bandit for choosing whom to audit, with an adaptive audit rate; collusion detection from answer correlations; adversaries that try to spot gold questions, and Sybil swarms; and real per-question model answers in place of the synthetic model. Code, tests and the script that regenerates every number are attached.

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